Mini-Project 02: Will This Be a Strong Sales Week?

Bayes' theorem and Naive Bayes on real Walmart sales data | In class (one session) | Pairs or individuals | You are the Data Analyst and the ML Engineer

How to read this brief. Section 1 is the situation you walk into. Section 2 is the data, and Section 3 defines the events you will measure. Section 4 is a warm-up by hand. Section 5 lists the tasks on the real data: first the probability of each event, then conditional probabilities, Bayes' theorem, and a Naive Bayes classifier. Project rules are labelled R1, R2, . . . ; tasks are labelled P (probability), B (Bayes) and N (Naive Bayes). Keywords are in red.

Item	Detail
Duration	One class session (about 2.5 hours); submit by 11:59 pm the same day
Team	Pairs (one Data Analyst, one ML Engineer) or individuals (both roles)
Method	Conditional probability, the law of total probability, Bayes' theorem, Naive Bayes
Data	Walmart Dataset (Kaggle): `Walmart.csv`, 6,435 rows, 45 stores $\times$ 143 weeks, February 2010 to October 2012: `https://www.kaggle.com/datasets/yasserh/walmart-dataset`
Submit	One notebook, the filled-in answer sheet (Section 7) and a 5-sentence note to the stakeholder

1. The Scenario

1.1 The Company and Your Team

DEFINITION

Walmart runs thousands of stores; this dataset covers 45 of them. Each store's weekly sales go up and down with holidays, the season, the weather and the local economy. You are on a retail analytics team that plans staffing and stock for these 45 stores. *(The people and requests in this scenario are fictional; the data is public.)*

EXAMPLE

Store 1 sold $1.64 million in the week of 5 February 2010 and $1.64 million again the next week, which was the Super Bowl holiday week. In the weeks of Thanksgiving and Christmas many stores sell far more than usual; in other weeks, sales move much less.

1.2 The Request from the Stakeholder

From: Maria Lopez, Regional Operations Manager **To:** Retail Analytics Team **Subject:** Extra staff: which weeks?

Hi team,

We add extra staff and stock in the weeks we expect to be busy. Right now we guess. I need numbers:

1. **How often** does a store have a *strong week*, and is it more likely in holiday weeks, in November–December, or in cold weeks?
2. Our current rule is "add extra staff only in holiday weeks". If it is a holiday week, how likely is a strong week really? And how many strong weeks does the rule miss?
3. Can a **simple model** tell us which weeks will be strong, better than the holiday rule?

Please answer in plain words, with probabilities. Use store numbers only, no store addresses or managers' names.

Thanks, Maria

1.3 What Success Looks Like

DEFINITION

You succeed if you can give Maria, with numbers: (1) the probability of each event in Section 3; (2) how the chance of a strong week changes *given* a holiday, the season, the weather and the store size; (3) the probability of a strong week in a holiday week, found with Bayes' theorem and checked by counting; (4) a Naive Bayes model, evaluated on weeks it never saw, compared with the holiday rule.

EXAMPLE

A measurable claim for the note: "A store has a strong week 12% of the time. In a holiday week it rises to 30%, but the holiday rule finds only 1 in 6 of all strong weeks." *(These numbers are illustrations; yours will differ.)*

Project rules

R1 Write down the exact definition of every event (Section 3) before computing anything. Use the same definitions in every task.

R2 One row = one store in one week (a store-week). The sample space S is all 6,435 store-weeks.

R3 For the Naive Bayes model (N tasks), train on 2010–2011 and test on 2012. Compute each store's average sales from the *training* weeks only, so the test weeks stay unseen.

R4 Use store numbers only in the note and the figures.

R5 Every probability you report must be checked a second way (by counting, by the complement, or by summing to 1).

2. The Data

DEFINITION

Download `Walmart.csv` from the Kaggle page in the table above. It has 8 columns and no missing values.

Column	Meaning	Watch out for
Store	store number, 1 to 45	a label, not a quantity
Date	the week (it ends on a Friday)	written day-month-year, e.g. `05-02-2010`: use `dayfirst=True`
Weekly_Sales	sales of the store that week (US dollars)	stores differ a lot in size: compare each store with its own average
Holiday_Flag	1 if the week contains a holiday, else 0	four holidays: Super Bowl, Labor Day, Thanksgiving, Christmas
Temperature	average temperature that week (°F)	
Fuel_Price	fuel cost in the region ($ per gallon)	
CPI	consumer price index	not used as an event here
Unemployment	unemployment rate in the region (%)	

EXAMPLE

Holiday weeks in the data (the `Date` of the holiday week): Super Bowl 12-Feb-2010, 11-Feb-2011, 10-Feb-2012; Labor Day 10-Sep-2010, 09-Sep-2011, 07-Sep-2012; Thanksgiving 26-Nov-2010, 25-Nov-2011; Christmas 31-Dec-2010, 30-Dec-2011. That is 10 holiday weeks for each of the 45 stores. The data ends in October 2012, so 2012 has no Thanksgiving or Christmas week.

3. The Events

DEFINITION

An event is a yes/no question about one randomly chosen store-week. These are the events for the whole project (R1).

Event	Definition (for one store-week)	Notation
S	a strong week: sales at least 10% above that store's average weekly sales	$P(S)$
H	a holiday week: `Holiday_Flag` $= 1$	$P(H)$
Q	the holiday season: the week is in November or December	$P(Q)$
C	a cold week: temperature below 40 °F	$P(C)$
F	expensive fuel: fuel price at least \$3.50	$P(F)$
J	high unemployment: unemployment at least 8%	$P(J)$
G	a large store: the store's average weekly sales are at least \$1,000,000	$P(G)$

EXAMPLE

Store 1 in the week of 12-Feb-2010: sales \$1,641,957, Super Bowl week, 38.5 °F, fuel \$2.55, unemployment 8.1%. It belongs to H, C and J, and (because Store 1 averages more than \$1 million a week) to G, and to the complements Q^c and F^c. Whether it is in S depends on Store 1's average, which you will compute.

4. Warm-up by Hand (20 minutes, no computer)

DEFINITION

Before using the real data, practise on a small *made-up* sample of 20 store-weeks (not from the file). Each store-week is described by three events: strong week (S), holiday week (H) and holiday season (Q).

	Holiday week (H)		**Not a holiday week** (H^c)		
	Nov–Dec (Q)	other months (Q^c)	Nov–Dec (Q)	other months (Q^c)	**Total**
Strong week (S)	2	1	2	1	6
Not strong (S^c)	0	1	1	12	14
Total	2	2	3	13	20

Warm-up questions (answer on paper; check with your partner)

W1. Probability of each event: find $P(S)$, $P(H)$ and $P(Q)$.

W2. Conditional: find $P(S \mid H)$ and $P(S \mid H^c)$. How much more likely is a strong week in a holiday week?

W3. Bayes' theorem: find $P(H \mid S)$ from $P(S \mid H)$, $P(H)$ and $P(S)$. Then check it by counting inside the 6 strong weeks.

W4. Total probability: rebuild $P(S)$ from $P(S \mid H)P(H) + P(S \mid H^c)P(H^c)$.

W5. Independence: is $P(S \cap H) = P(S)P(H)$? What does your answer mean?

W6. The rule "extra staff only in holiday weeks": find its precision $P(S \mid H)$ and its recall $P(H \mid S)$. What fraction of strong weeks does it miss?

W7. Naive Bayes by hand: for a store-week that is a holiday week *and* in Nov–Dec, compute the two scores $P(S)P(H \mid S)P(Q \mid S)$ and $P(S^c)P(H \mid S^c)P(Q \mid S^c)$, and normalize them. Do the same for a week that is neither. Compare each with the true fraction of strong weeks in that column of the table.

5. Tasks on the Real Data

5.1 Part P — The probability of each event

Task	What to do	Output
P1	Load `Walmart.csv`, parse the dates (`dayfirst=True`), and check: 6,435 rows, 45 stores, 143 weeks, and 450 holiday store-weeks.	Data check
P2	Add the seven event columns of Section 3. Compute the probability of each event S, H, Q, C, F, J, G by counting: $P(E) = \#E/\#S$. Check each with its complement (R5). (Your $P(H)$ should be $450/6435 = 0.070$.)	Event table
P3	Compute the joint probabilities $P(S \cap H)$, $P(S \cap Q)$, $P(H \cap Q)$, $P(S \cap C)$.	Joint table

5.2 Part B — Conditional probability and Bayes' theorem

Task	What to do	Output
B1	**Holidays.** Compute $P(S \mid H)$ and $P(S \mid H^c)$ by counting inside each group.	Two numbers, one sentence
B2	**Reverse it with Bayes.** Compute $P(H \mid S) = \dfrac{P(S \mid H)P(H)}{P(S)}$, then check it by counting inside the strong weeks.	Both values, equal
B3	**Total probability over the holidays.** Split the store-weeks into five groups: Super Bowl, Labor Day, Thanksgiving, Christmas, no holiday. Compute $P(\text{group})$ and $P(S \mid \text{group})$ for each, and rebuild $P(S) = \sum_g P(S \mid g)P(g)$. Which holiday has the highest $P(S \mid g)$? Which group gives the largest share of all strong weeks, $P(g \mid S)$?	Table with 5 rows
B4	**Season.** Compute $P(S \mid Q)$ and $P(Q \mid S)$. Explain in words why they are different.	Two numbers, one sentence
B5	**The staffing rule** "extra staff only in holiday weeks". Compute precision $P(S \mid H)$ with Bayes' theorem from $P(H \mid S)$, $P(H \mid S^c)$ and $P(S)$, and check by counting. Compute recall $P(H \mid S)$: what fraction of strong weeks get extra staff? Repeat for the rule "extra staff in November–December" (Q) and compare.	Precision and recall, two rules
B6	**Base rate.** Using your $P(H \mid S)$ and $P(H \mid S^c)$ from B5, what would the precision of the holiday rule be if strong weeks were twice as common? Half as common?	Two numbers
B7	**Independence checks.** For the pairs (S, H), (S, Q), (S, C), (S, F), (S, J), (S, G): compare $P(A \cap B)$ with $P(A)P(B)$. Which events carry information about strong weeks?	Table, one sentence

5.3 Part N — A Naive Bayes classifier for strong weeks

Task	What to do	Output
N1	**Split by time (R3).** Training set: 2010–2011. Test set: 2012. Recompute each store's average from the training weeks only, then the events S and G. Report $P(S)$ in each set.	Two numbers
N2	**Train by counting.** On the training set only, compute the priors $P(S)$, $P(S^c)$ and the likelihood tables $P(\text{feature} \mid S)$ and $P(\text{feature} \mid S^c)$ for the six features H, Q, C, F, J, G.	Likelihood tables
N3	**Predict by hand.** Compute $P(S \mid \boldsymbol{x})$ for two profiles: (a) holiday week, Nov–Dec, cold, cheap fuel, low unemployment, large store; (b) no holiday, not Nov–Dec, not cold, expensive fuel, high unemployment, small store. Show the two scores and the normalization.	Two posteriors
N4	**Check the naive assumption.** For the pair (H, Q), compare $P(H \cap Q \mid S)$ with $P(H \mid S)P(Q \mid S)$ in the training set. Why are these two features *not* independent?	One comparison
N5	**scikit-learn.** Fit `CategoricalNB` on the training set with the six 0/1 features. Check that `predict_proba` for profiles (a) and (b) matches N3.	Matching numbers
N6	**Evaluate on the test set.** Report the confusion matrix, accuracy, precision $P(S \mid \hat{S})$ and recall $P(\hat{S} \mid S)$. Compare with the baselines "never strong" and "strong if holiday week". The test year has no Thanksgiving or Christmas: how does that affect your results?	Results table
N7	**Threshold (stretch).** Predict S when $P(S \mid \boldsymbol{x}) > t$ for $t = 0.1, 0.2, 0.3, 0.5$. How do precision and recall change?	Short table

5.4 Joint task

J1	A 5-sentence note to Maria in plain words: how often stores have a strong week; which holiday and season matter most; the honest truth about the holiday rule (precision and what it misses); whether the model beats the rule; one risk.	note.txt

6. Starter Code

DEFINITION

These pieces get you moving. They are not a complete solution.

```
import numpy as np, pandas as pd
from sklearn.naive_bayes import CategoricalNB
from sklearn.metrics import confusion_matrix, precision_score, recall_score

df = pd.read_csv("data/Walmart.csv")
df["Date"] = pd.to_datetime(df["Date"], dayfirst=True)
print(len(df), df["Store"].nunique(), df["Date"].nunique(), df["Holiday_Flag"].sum())

def add_events(d, store_avg):
    d = d.copy()
    avg = d["Store"].map(store_avg) # each store's own average
    d["S"] = d["Weekly_Sales"] >= 1.10 * avg # strong week
    d["H"] = d["Holiday_Flag"] == 1
    d["Q"] = d["Date"].dt.month.isin([11, 12])
    d["C"] = d["Temperature"] < 40
    d["F"] = d["Fuel_Price"] >= 3.50
    d["J"] = d["Unemployment"] >= 8
    d["G"] = avg >= 1_000_000 # large store
    return d

# P2: probability of each event on all store-weeks, with the complement check (R5)
```

```
ev = add_events(df, df.groupby("Store")["Weekly_Sales"].mean())
for e in "SHQCFJG":
    p = ev[e].mean(); print(e, round(p, 4), "check:", round(p + (1 - ev[e]).mean(), 4))

# B1 and B2: conditional probability and Bayes
p_S_given_H = ev.loc[ev.H, "S"].mean()
p_H_given_S = p_S_given_H * ev.H.mean() / ev.S.mean() # Bayes
print(p_H_given_S, ev.loc[ev.S, "H"].mean()) # same number (R5)

# N1 and N5: time split; store averages from the training years only (R3)
train_raw, test_raw = df[df.Date.dt.year <= 2011], df[df.Date.dt.year == 2012]
store_avg = train_raw.groupby("Store")["Weekly_Sales"].mean()
train, test = add_events(train_raw, store_avg), add_events(test_raw, store_avg)
X = ["H", "Q", "C", "F", "J", "G"]
nb = CategoricalNB(alpha=1).fit(train[X].astype(int), train["S"])
pred = nb.predict(test[X].astype(int))
print(confusion_matrix(test["S"], pred))
```

7. Answer Sheet (fill in and submit)

Task	Quantity	Your value
P1	rows, stores, weeks, holiday store-weeks	
P2	$P(S),\ P(H),\ P(Q),\ P(C),\ P(F),\ P(J),\ P(G)$	
P3	$P(S \cap H),\ P(S \cap Q),\ P(H \cap Q),\ P(S \cap C)$	
B1	$P(S \mid H),\ P(S \mid H^c)$	
B2	$P(H \mid S)$ by Bayes, and by counting	
B3	holiday with the highest $P(S \mid g)$; group with the largest $P(g \mid S)$	
B4	$P(S \mid Q),\ P(Q \mid S)$	
B5	holiday rule: precision, recall; Nov–Dec rule: precision, recall	
B6	precision of the holiday rule if strong weeks doubled, if halved	
N3	$P(S \mid$ profile a$),\ P(S \mid$ profile b$)$	
N6	accuracy, precision, recall: Naive Bayes / “never strong” / “strong if holiday”	

8. Grading (100 points)

Criterion	Points	Full marks look like
Warm-up by hand (W1–W7)	15	All seven answered with steps; W7 compares Naive Bayes with the true fractions
Probability of each event and joint probabilities (P1–P3)	15	Data checks pass; every probability checked with its complement
Conditional probability and Bayes (B1–B7)	30	Bayes results checked by counting; total probability rebuilds $P(S)$; both staffing rules compared
Naive Bayes (N1–N7)	25	Time split respected (store averages from training only); hand calculation matches scikit-learn; honest comparison with both baselines
Note to the stakeholder (J1)	10	Five plain sentences with numbers, including what the holiday rule misses
Code quality and rules (R1–R5)	5	Definitions written first; notebook runs top to bottom